Chapter 39

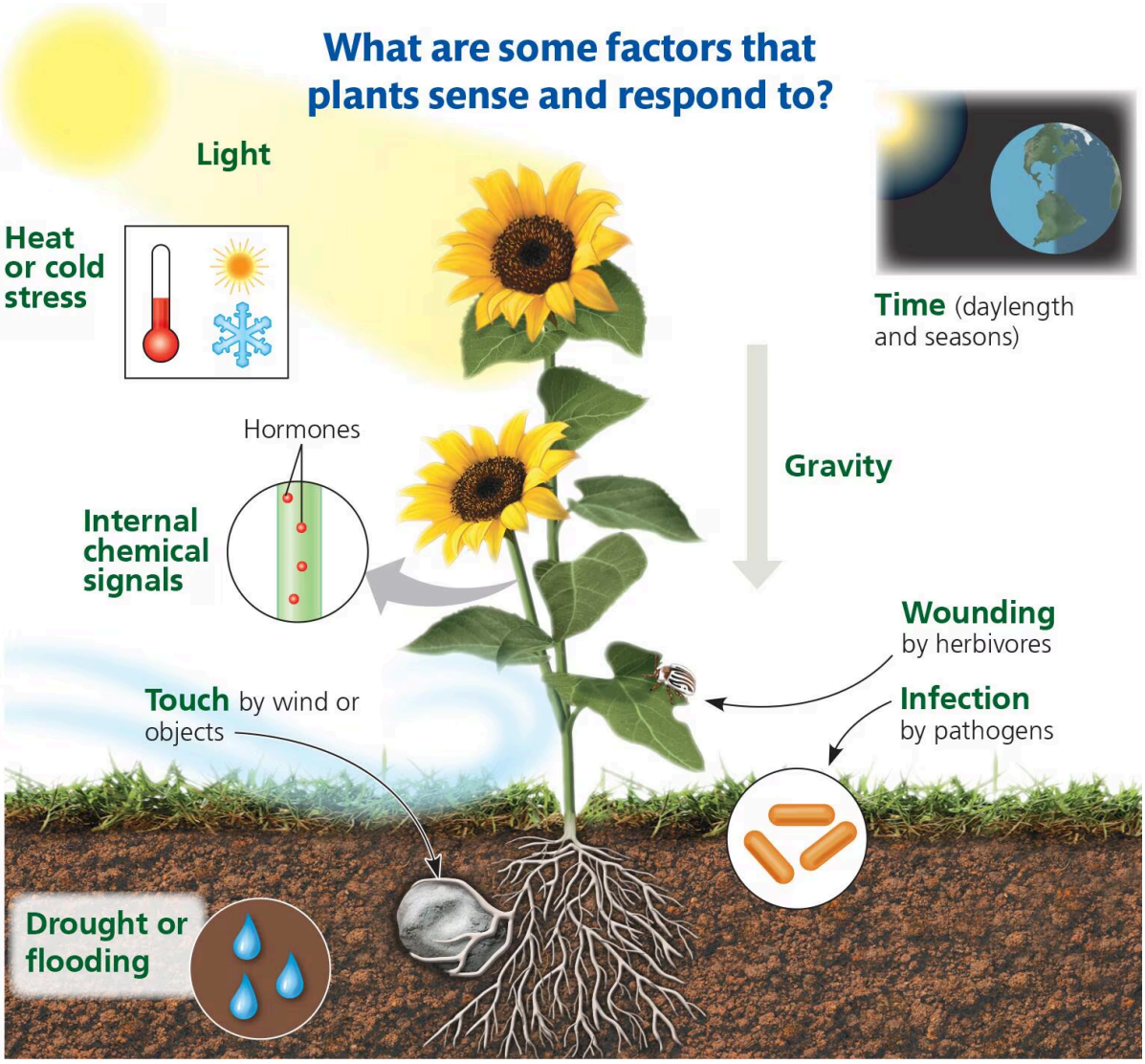
Campbell Biology, AP Edition, 12e

Chapter 39

Plant Responses to Internal and External Signals

Figure 39.1

Sunflowers track the sun from east to west each day. After sunset, they reverse direction, facing the direction of the next sunrise. By facing the hot sun during the day, the floral heads become warmer and release greater amounts of chemicals that attract pollinators. Light is just one of the many factors to which a plant responds.



An illustration depicts the stimuli that plants sense and respond to. Some factors are heat or cold stress (depicted by a thermometer), light (depicted by the sun), gravity (depicted by an arrow pointing towards the ground), time of day and season (depicted by the sun shining on a part of the Earth), internal chemical signals (depicted by globular hormones inside the plant), wounding by herbivores, infection by pathogens (depicted by images of bacteria), touch by wind or objects, and drought of flooding (depicted by water droplets).

Key Concepts

- **39.1Signal transduction pathways link signal reception to response****
- **39.2*Plants use chemicals to communicate*****
- **39.3Responses to light are critical for plant success****
- **39.4Plants respond to a wide variety of stimuli other than light****

- **39.5Plants respond to attacks by pathogens and herbivores****

In plants, physiological events involve the interaction between environmental signals and internal molecular signals **(BIG IDEA 2)**. Plants' adaptations for cellular communication **(BIG IDEA 3)** enable them to respond to seasonal changes that can affect when and how they interact with other organisms in their environment **(BIG IDEA 4)**.

Study Tip

Make a table: As you read the chapter, add specific examples for each of the general categories of responses shown in the diagram.

Factor	Example of plant response
Light	Seed germination in response to red light

Get Ready for This Chapter

In this chapter, you will learn how plants have mechanisms for sensing abiotic and biotic changes in their environment, integrating this information, and responding appropriately, often by altering their growth pattern. Before beginning this chapter, you should be able to do the following:

- Describe how scientists study light-dependent processes (see Concept 10.2 and Figure 10.9).
- Describe the basic steps of signal transduction (see Concept 11.3).
- Describe the structure of the plant cell wall (see Concept 6.7 and Figure 6.27).

The following questions provide a quick check of your basic knowledge in these areas.

1. An action spectrum is best described as

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the amount of energy that reactants must absorb before a chemical reaction can
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That's incorrect. Try again.

Incorrect. Review the following material: Physiologists who study how organisms respond to light will often create an action spectrum, a graph of how a particular light-sensitive process is affected by different wavelengths of light.

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a graph of the absorption of light by isolated molecules.
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That's incorrect. Try again.

Incorrect. Review the following material: Physiologists who study how organisms respond to light will often create an action spectrum, a graph of how a particular light-sensitive process is affected by different wavelengths of light.

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a propagating electrical signal.
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That's incorrect. Try again.

Incorrect. Review the following material: Physiologists who study how organisms respond to light will often create an action spectrum, a graph of how a particular light-sensitive process is affected by different wavelengths of light.

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a graph of the relative effectiveness of different wavelengths of radiation in driv
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That's correct.

Correct! If you had trouble with this question, review the following material: Physiologists who study how organisms respond to light will often create an action spectrum, a graph of how a particular light-sensitive process is affected by different wavelengths of light.

2. The term *second messengers* refers to

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small, water-soluble, non-protein molecules produced or released intracellular
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That's correct.

Correct! If you had trouble with this question, review the following material: Cascades of molecular interactions relay signals from receptors to target molecules in the cell. Second messengers relay a signal to a cell’s interior in response to a signal interacting with a cell receptor.

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chemical messengers of secondary importance.
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That's incorrect. Try again.

Incorrect. Review the following material: Cascades of molecular interactions relay signals from receptors to target molecules in the cell. Second messengers relay a signal to a cell’s interior in response to a signal interacting with a cell receptor.

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chemical messengers that are transported throughout the organism.
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That's incorrect. Try again.

Incorrect. Review the following material: Cascades of molecular interactions relay signals from receptors to target molecules in the cell. Second messengers relay a signal to a cell's interior in response to a signal interacting with a cell receptor.

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chemical messengers that evolved after primary messengers.
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That's incorrect. Try again.

Incorrect. Review the following material: Cascades of molecular interactions relay signals from receptors to target molecules in the cell. Second messengers relay a signal to a cell's interior in response to a signal interacting with a cell receptor.

3. Which one of the following statements about cell walls is true?

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The plant cell wall is inflexible.
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That's incorrect. Try again.

Incorrect. Review the following material: Plant cell walls prevent the plant protoplast from lysing due to an excessive uptake of water by osmosis. The pushing of the protoplast against the cell wall is called turgor pressure.

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The plant cell wall prevents excessive uptake of water.
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That's correct.

Correct! If you had trouble with this question, review the following material: Plant cell walls prevent the plant protoplast from lysing due to an excessive uptake of water by osmosis. The pushing of the protoplast against the cell wall is called turgor pressure.

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The plant cell wall is made exclusively of cellulose.
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That's incorrect. Try again.

Incorrect. Review the following material: Plant cell walls prevent the plant protoplast from lysing due to an excessive uptake of water by osmosis. The pushing of the protoplast against the cell wall is called turgor pressure.

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        Plant cells are unique in having cell walls.
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That's incorrect. Try again.

Incorrect. Review the following material: Plant cell walls prevent the plant protoplast from lysing due to an excessive uptake of water by osmosis. The pushing of the protoplast against the cell wall is called turgor pressure.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 39.1Trace the steps of a signal transduction pathway in a plant.
- 39.2Identify plant hormones and their functions.
- 39.3Describe plant responses to light.
- 39.4Give examples that show how plants respond to stimuli.
- 39.5Describe plant defenses against herbivores and pathogens.